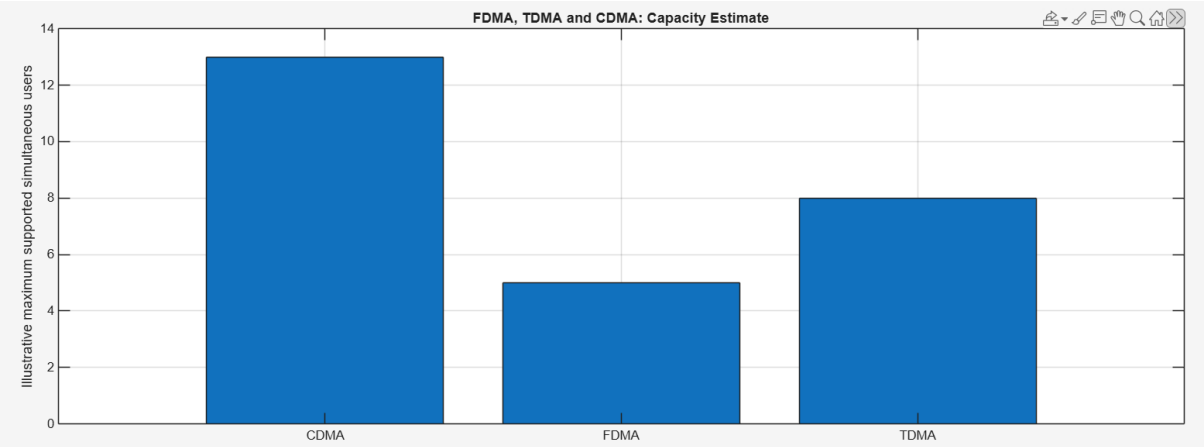
```
clear; clc; close all;
totalBandwidthHz = 1e6;
fdmaChannelHz = 200e3;
tdmaSlots = 8;
processingGain = 64;
targetSIRdB = 7;
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB / 10);
cdmaUsers = floor(1 + processingGain / targetSIR);
fprintf('Illustrative maximum supported users\n');
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
    cdmaUsers, targetSIRdB);
K = 2:30;
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];
results = table(method, supportedUsers);
disp(results);
figure;
bar(categorical(method), supportedUsers);
ylabel('Illustrative maximum supported simultaneous users');
title('FDMA, TDMA and CDMA: Capacity Estimate');
grid on;
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];
system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];
accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; "OFDMA"; "OFDMA-based"];
keyIdea = ["Separate frequency channels"; ...
    "Separate time slots"; ...
    "Code-based user separation"; ...
    "Wideband code-based access"; ...
    "Orthogonal frequency resources"; ...
    "Flexible OFDM-based radio access"];
evolutionTable = table(generation, system, accessMethod, keyIdea);
disp(evolutionTable);
figure;
tiledlayout(1,3);
nexttile;
hold on;
colors = lines(4);
for u = 1:4
    rectangle('Position',[0, u-1, 1, 0.8], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
    text(0.5, u-0.6, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 1]); ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');
set(gca,'YTick',[]);
hold off;
nexttile;
hold on;
for u = 1:4
    rectangle('Position',[u-1, 0, 0.8, 1], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
    text(u-0.6, 0.5, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 4]); ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca,'YTick',[]);
hold off;
nexttile;
hold on;
```

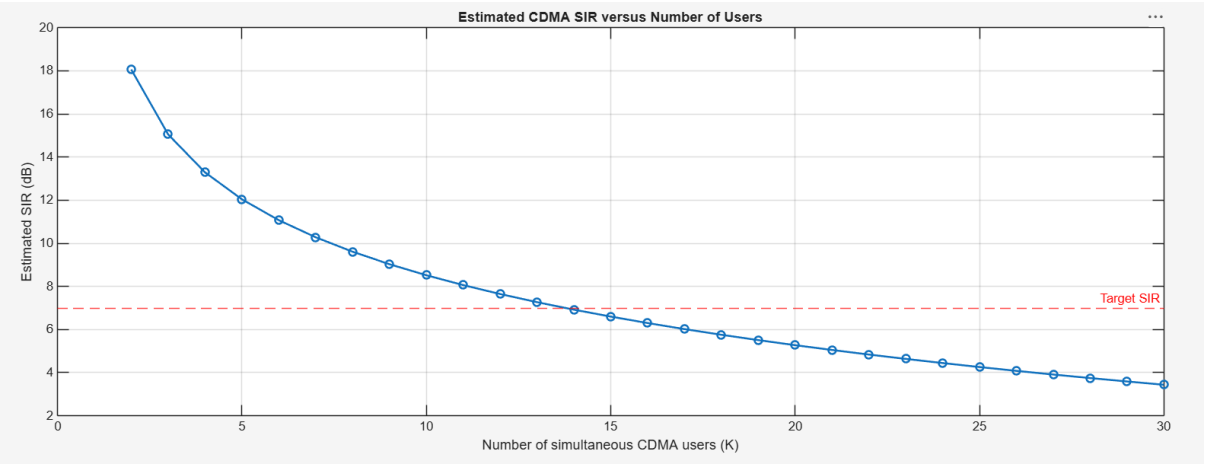
```
    rectangle('Position',[0, 0, 1, 1], ...
        'FaceColor',colors(u,:), 'FaceAlpha',0.22, ...
        'EdgeColor',colors(u,:), 'LineWidth',1.5);
end
text(0.5, 0.62, 'Users 1-4 share', 'HorizontalAlignment','center');
text(0.5, 0.40, 'band; different codes', 'HorizontalAlignment','center');
xlim([0 1]); ylim([0 1]);
xlabel('Time');
ylabel('Frequency band');
title('CDMA');
set(gca,'YTick',[]);
hold off;
sgtitle('Conceptual Time-Frequency Resource Allocation');
```

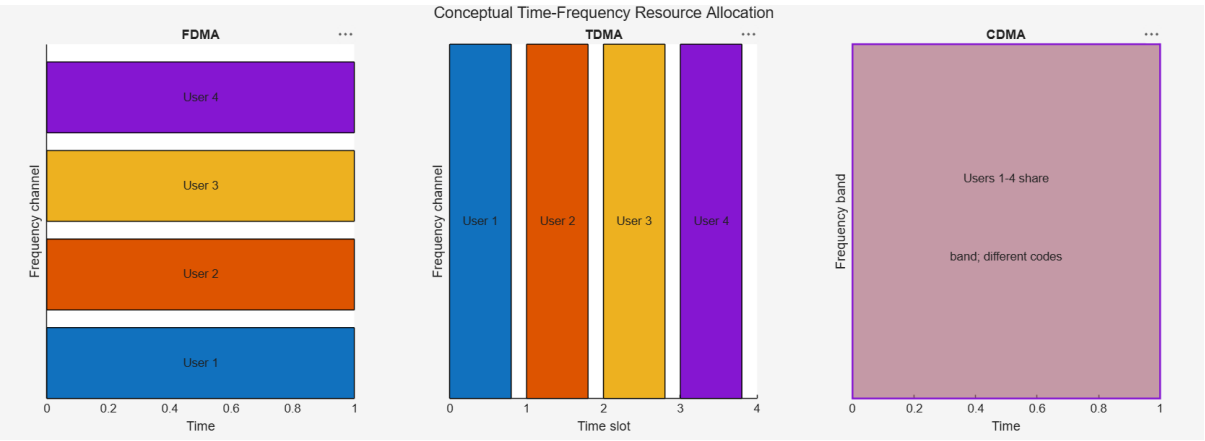
```
Illustrative maximum supported users
FDMA: 5 users
TDMA: 8 users
CDMA: 13 users at a target SIR of 7.0 dB
```

method	supportedUsers
"FDMA"	5
"TDMA"	8
"CDMA"	13

generation	system	accessMethod	keyIdea
"1G"	"AMPS"	"FDMA"	"Separate frequency channels"
"2G"	"GSM"	"TDMA"	"Separate time slots"
"2G"	"IS-95"	"CDMA"	"Code-based user separation"
"3G"	"UMTS"	"WCDMA"	"Wideband code-based access"
"4G"	"LTE"	"OFDMA"	"Orthogonal frequency resources"
"5G"	"5G NR"	"OFDMA-based"	"Flexible OFDM-based radio access"




Estimated CDMA SIR versus Number of Users
Estimated SIR (dB)
Number of simultaneous CDMA users (K)
Target SIR
20
18
16
14
12
10
8
6
4
2
0
5
10
15
20
25
30


Conceptual Time-Frequency Resource Allocation
FDMA
Frequency channel
User 4
User 3
User 2
User 1
Time
TDMA
Frequency channel
User 1
User 2
User 3
User 4
Time slot
CDMA
Frequency band
Users 1-4 share
band; different codes
Time